

Advances in Science and Technology Through Frugality

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Abstract—Sustainable development necessitates the management of progress in science and technology for society's betterment while preserving Earth's resources. In this regard, the appearance in recent years of low-cost sophisticated products consuming resources economically is a force for good. This effort terms each of these products as an advanced frugal innovation to highlight the frugality in resource consumption during the realization of these innovations through advances in various scientific disciplines. Other than 25 examples of advanced frugal innovations in a wide range of sectors from rural electrification to particle physics, a framework for the systematic realization of such innovations for sustainable development has also been described. The innovations showcased bring out the use of advances in science and technology or cutting edge knowledge for creating these low-cost sophisticated products. Moreover, the need for innovators typically with training in advanced knowledge areas to handle the sophistication in technology for the successful fruition of these innovations is also discussed. This effort supports the frugal design and frugal engineering of innovations, whether grassroots or the advanced type, through sound scientific principles for the creation of robust products, especially where the human life is at stake, in various sectors for all-round sustainable development.

Key words: Advanced frugal innovation, sustainable development, low cost philosophy, science, technology

1. INTRODUCTION

THE limits on planetary boundaries (Rockström *et al.* 2009) required in this age of the *Anthropocene* necessitate curbs on numerous human activities for mitigating crises such as *climate change* and *resource scarcity*, to name a few. In this regard, adopting sustainability principles for all round sustainable development would help achieve better standards of living while leaving a functioning earth to posterity (Arnold 2015). In fact, innovations for *sustainable development* have been suggested to be a win-win situation against a background of environmental and also economic upheaval (Boons *et al.* 2013). Therefore, *sustainable development* is here to stay from this century onwards to facilitate the habitation of this planet against the forces of planetary crises (Sachs 2015). In this regard, *science &*

technology (S&T) based innovations achieved through frugal means can go a long way in conserving material resources and thus abet *sustainable development*. The advent of *frugal-innovation* over the past decade has been observed to be a grassroots phenomenon, where scarcity in resources dictates the frugality in innovations (Gupta 2012, Martin and Upham 2015). The term "resources" covers raw materials, knowledge, skilled personnel and other relevant tangible and intangible inputs required for the successful realization of these innovations. Therefore grassroots innovations provide a rich trove of knowledge on the effective utilization of meager resources to the advantage of people at the bottom of the pyramid (Gupta 2013). Other than *Jugaad* (The New York Times 2010), the quintessential grassroots innovation, examples of *frugal-innovation* include an

earthen-refrigerator (Entrepreneur 2010) and the bamboo-microscope (Basu 2007), to name a few. However, in recent years, frugality has also been observed in several innovations made by following the advances in S&T, ranging from cars to healthcare, astronomy and particle physics (Rao 2013, Rosca *et al.* 2016). Although these innovations have been realized with an eye for lowering costs, the basic principle of restraining resources in achieving their aim gel well with the concept of their grassroots brethren.

Therefore, the origin of *frugal-innovation* as a grassroots concept has not precluded its application to sophisticated technology albeit in a reversed manner. In other words, innovations at a grassroots level are sought to overcome the challenging constraints of scarcity thereby leading to a product or service that is affordable and which functions with minimal features. In contrast, numerous sophisticated innovations are being realized by both industry and academia with an eye for lowering the costs and hence the price of the product. The motivation for minimizing the cost has been, not least, due to: affordability by society at large (Technology Review 2011, Joyce 2012); budgetary crunch in science and engineering (Downer and Zgadzaj 2014) in the wake of manmade crises such as the financial one in 2008–2009 (World Bank 2009). Consequently, costs are tightly controlled by economizing the use of resources for achieving a product with good functionality. As a result, these sophisticated low-cost innovations chase frugality as opposed to a grassroots product being born under conditions of scarcity.

2. ADVANCED FRUGAL INNOVATION (AFI)

This effort terms such a sophisticated low-cost product as an *Advanced*

Frugal Innovation (AFI) since it leverages advances in S&T to capture the frugality inherent in a grassroots *frugal innovation*. Therefore an Advanced Frugal Innovation (AFI) has the following defining features:

- a. Their cost is low due to their frugal nature.
- b. They leverage advances in S&T from the incipient stage. Science and technology, as used in this effort, encompasses the use of principles and/or advances in sciences, engineering, technology, mathematics and any other knowledge-based area relevant to a given AFI. Unlike their grassroots brethren that depend on serendipity and/or tinkering for creation (Gupta 2013), AFIs need significant involvement of S&T from the incipient stage to account for the sophistication in their designs.
- c. They are undertaken by workers with advanced degrees - see Section 4.1 - who can leverage advances in S&T for achieving an AFI.

3. RESULTS: EXAMPLES OF ADVANCED FRUGAL INNOVATIONS

Table 1 lists 25 examples of *advanced frugal innovations* that have appeared in the past decade in sectors spanning from cars to healthcare, weaponry and high-energy-particle-physics. Although all of them share the primary characteristic of being cheap, a majority of these are also compact with a no-frills structure as evidenced by column 5 of Table 1. The AFIs with no entries under “Compact Design” are deemed so due to the implausibility of creating a compact version with Sandia’s robotic hand being an obvious example. Overall, these examples clearly attest to doing good science and engineering with less thereby eventually lowering the costs involved. The brief descriptions

to follow for some of these innovations mention their entry numbers in Table 1 in square parenthesis.

The rainfall sensor (Downer and Zgadzaj 2014) and plasma-collider (Austen 2015) utilize newer low-cost methods backed by research such as lightning data for rainfall estimation (Tollefson 2013) and a 30 cm plasma afterburner for boosting electrons coming from a conventional accelerator (Downer and Zgadzaj 2014), respectively. Traditionally, rainfall prediction uses expensive and bulky Doppler radar while the particle colliders are costly behemoths smashing particles that are accelerated over distances spanning several kilometers.

Continuing with plasma, Pavlovich (2014) has reported the production of nitrogen dioxide (NO₂) and nitric oxide (NO) species in the air through inexpensive spark-like plasma technology that runs on low cost power supplies (Norton 2006). The NO₂ and NO species have antimicrobial properties that make this *frugal plasma* technology apt for applications involving sterilization and antiseptis, to name a few.

The frugal internal combustion engine (Entrepreneur 2010) is another instance of research used for the creation of advanced turbochargers that deliver more power and torque from a smaller and hence cheaper, four cylinder engine, while also economizing on both fuel and pollution. Such puny but powerful engines are used in the luxury car models being rolled out of the US and Europe (The Economist 2015).

Smart-mobile-phony has been used for the low cost and successful screening of communicable and non-communicable diseases such as tuberculosis (Martin and Upham 2015) and intestinal worms (Basu 2007). These applications use probes

and/or inexpensive lenses in conjunction with mobile phony for the detection of pathogens (Joyce 2012, 2013b). Some strains of Ebola (Ball

2001) have been detected through dehydrated gene circuits laid on a paper. When hydrated these circuits have been successful in detecting

Ebola strains such as Sudan and Zaire (Nature 2014a). Recently, a low-cost palm-sized DNA sequencer (Nordling 2012) has been developed

Table 1 Advanced Frugal Innovations in Science & Technology

No	Advanced Frugal Innovation (AFI)	Manufacturer & Country	Frugal Features	Compact Design	Price
1	Light (http://gravitylight.org/)	GravityLight (GravityLight™, UK) (US & India)	Using the descent of a weight under gravity to generate 25 minutes of lighting without electricity	Yes	£3
2	Rural Electrification (Jeff 2014)		Uses cheap microgrids and smart meters to deliver eco-friendly electricity to rural India at a lower cost	–	–
3	Car (The Economist 2010)	Tata Motors (Tata Nano®, India)	Basic “no frills” car that was built to minimize unnecessary features	Yes	\$2200
4	Frugal Four Cylinder Engine (The Economist 2015)	(USA and Europe)	Use of advanced turbochargers for amplifying the power and torque of four cylinder engines while providing economy in both fuel and pollution	Yes	–
5	Electrocardiogram (Immelt <i>et al.</i> 2009, The Economist 2010)	General Electric (Mac 400® & Mac 800®, India)	ECG reduced to a handheld device with a built in small printer	Yes	\$800
6	Ultrasound (Immelt <i>et al.</i> 2009)	General Electric (China)	Portable machine consisting of a laptop computer with sophisticated software and associated probe	Yes	\$15 000 to \$100 000
7	Speech synthesizer [MIT (Technology Review 2011)]	Invention Labs (Avaz®, India)	Compact unit handling multiple languages made cheap by working directly with Asian hardware manufacturers	Yes	\$800
8	TB Probe (Joyce 2012)	Stanford University (US)	Probe fluorescent molecule when cleaved by a TB protein releases faint blue color picked up by a smartphone camera and later transmitted to clinicians for frugal diagnosis	Yes	–
9	Identification of Intestinal Worms (Nature 2013)	Toronto General Hospital (Canada)	An inexpensive lens attached to an <i>iPhone</i> allows low cost optical microscopy of stool samples	Yes	–
10	Identification of Ebola (Nature 2014a)	Boston University (USA)	Gene circuits on paper for identifying pathogens	Yes	–
11	Frugal Cancer Detection (Wired 2014)	Miroculus (Miriam, US)	Monitors apt microRNA in blood sample through a compact 3D printed device that communicates with the Cloud	Yes	–
12	Portable DNA Sequencer (Hayden 2015)	Oxford Nanopore Technologies (MinION, UK)	Palm sized gene sequencer that plugs into a laptop USB for displaying data online instead of at end of run	Yes	–
13	Robotic Hand (Greenemeier 2012)	Sandia Labs (US)	Improved dexterity mimicking real time gestures through an electronic system comprised of cell-phone-parts	–	\$10 000
14	Swarms of Simpler Robots (Nature 2014b)	Massachusetts Institute of Technology (Kilobots, US)	Swarms of low-cost simple robots perform complex tasks in unison	Yes	–
15	Robotic Submarine (Davis 2014)	Falcon Robotics (Stinky, US)	Underwater robotic submarine fabricated under a shoe-string budget wins NASA sponsored Marine Advanced Technology Education Robotics Competition	Yes	\$1000
16	Pollution sensors (Austen 2015)	Do-It-Yourself (DIY) Kit (DustDuino; Air Quality Egg & Smart Citizen Kit, US)	Cheap sensors built through the DIY movement that provides instructions on the internet	Yes	\$185 (Air Quality Egg) \$50 (DustDuino)
17	Lightning sensor (Tollefson 2013)	Earth Networks (USA)	Lightning data tracked by sensors on top of mobile towers are converted to rainfall estimates by intricate software	–	\$1 000 000
18	Missile (The Economist 2012)	Naval Air Weapons Station (Spike, US)	Low-cost small missile made from off the shelf components. This has an ingenious optical guidance system.	Yes	\$5000
19	African Radio Telescope (Nordling 2012a)	(Square Kilometer Array, South Africa)	Satellite dishes used for telecommunications are retrofitted to create low-cost radio telescopes	–	R15 000 000 to 20 000 000
20	African Telescope for Spectroscopy (Nordling 2013b)	(South African Large Telescope (SALT), South Africa)	Design induced by a shoestring budget to build a telescope with minimal features for accurate spectroscopy	–	\$30 000 000
21	Indian Martian Spacecraft (The Economist 2014, Kumar 2014)	Indian Space Research Organization (Mangalyaan, India)	The spacecraft achieves its frugality through launch by a low-power rocket and also by carrying fewer sensors and instruments	Yes	\$74 000 000
22	Smaller satellites (Jones 2014)	(CubeSats, US)	Small and compact satellites comprised of cheap parts that are also launched free of cost	Yes	< \$100 000
23	Reuse of Rockets for Space Flight (Scientific American 2015)	SpaceX (Falcon 9, US)	Failed attempt on attempting to land safely and hence reuse, to reduce launching costs, the first stage Falcon 9 booster rocket	–	–
24	Plasma Particle Accelerator (Downer and Zgadzaj 2014)	SLAC National Accelerator Laboratory (FACET, US)	Use of plasma afterburner about 30 cm long has resulted in a low-cost technology capable of boosting electrons from a conventional accelerator 500 times	Yes	–
25	Frugal Plasma Technology (Pavlovich 2014)	University of California, Berkeley (Frugal Plasma, US)	Use of inexpensive spark-like plasma technology to create antimicrobial species in air for applications involving sterilization	–	–

The reference for the data appearing in these columns, except “Frugal Features” and “Compact Design”, is listed for each of the entries under the column on “Advanced Frugal Innovation (AFI)”.

whose coupling with mobile phony in the foreseeable future would allow fast identification of unknown *fauna* (Hayden 2015). All of these mobile phony based AFIs are compact when compared to the bigger setups housed in clinics with the supporting paraphernalia.

Cloud technology has been utilized by a group of entrepreneurs in communicating data with a low cost sensor fabricated by 3D printing (Joyce 2012). The sensor called *Miriam* analyses blood samples looking for suitable microRNA markers for cancer that are subsequently communicated to the biochemistry-based algorithms housed in the *Cloud* that aid in the fast and cheap but smart detection of different types of cancers (Wired 2014). The use of a cheap 3D printed sensor together with the *Cloud* technology make this application frugal.

The proliferation of *do it yourself* (DIY) kits has unleashed cheaper sensors (Nordling 2013) for tackling pollution that are being deployed by multitudes of mostly amateur scientists. The big-data streaming from myriads of these sensors has facilitated monitoring of pollution particulate that are $\geq 2.5 \mu\text{m}$ (Austen 2015).

4. DISCUSSION

The innovations in Table 1 were either developed as a new product, such as entries [1 to 12, 14, 16 to 18, 20 to 25], or were retrofitted from existing designs and/or the recycling of end-of-life components as is evident in the fabrication of the remaining AFIs including the *Square Kilometer Array* for radio telescope (Nordling 2012, Jeff 2014), improvised submarine *Stinky* (Nature 2013, Davis 2014), and the robotic hand (Greenemeier 2012). Notwithstanding their fabrication, these innovations share the traits of an AFI: realized by the

application of S&T; workers are cognizant of S&T principles through formal training and they are frugal.

4.1. Developers of Advanced Frugal Innovations

All of the innovations, except [1, 2, 15, 16, 22] in Table 1, were realized by workers with a strong background in science and technology with relevant details to be found in the references attached to each table entry. Even entries (Boons *et al.* 2013, Nordling 2013, Jones 2014) involve mentoring or development of the original AFIs for DIY practitioners by workers with advanced degrees. This is because these innovations have been realized by *research and development* (R&D) teams, in academia and/or industry, where the knowledge-base of the workforce needed to develop cars, internal-combustion engines, accelerators, missiles and rockets, to name some of the innovations listed in Table 1, is accrued typically through advanced degree programs beyond a bachelor's degree. Also, end users as reported by von Hippel (1976) were not the innovators for the AFIs listed in Table 1. However von Hippel's observation applies to *frugal innovations* of the grassroots type, which are typically makeshift arrangements arrived at by the ingenuity of end-users, as innovators, at the base of the pyramid (Gupta, 2012, 2013).

4.2. Research & Development on Advanced Frugal Innovations

As evidenced by the frugal features in Table 1, budgetary constraints have encouraged researchers to seek novel solutions that are also advancing the fields of science and engineering. The examples of the plasma collider; palm-sized DNA sequencer; and lightning sensor, to name a few, point to new science & technology expanding the frontiers of knowledge. Consequently, the conceptualization and realization of AFIs can benefit immensely from

research and development programs dedicated to their creation. This is because an effective R&D program will aid in the systematic application of advances in S&T to realize frugal innovations of the advanced type. Furthermore, such research activity could also aid in advancing S&T.

4.3. Design Philosophy for Advanced Frugal Innovations

The continued use of frugality to create AFIs warrants a philosophy rooted in science & technology that will aid in the methodical realization of more-with-less. A methodical approach is needed to improve the safety and performance of designs done-with-less, i.e. realized with lesser resources. Such a philosophy is also required to avoid misuse of the frugality approach by, not least, profit-seeking entities taking it up in a haphazard manner. Moreover, this philosophy has to apply to the creation of AFIs as a novelty from scratch and/or the assembly of components salvaged from end-of-life systems.

The three essential elements of a low-cost S&T philosophy are the following. First, the emerging area of frugal-design is interdisciplinary which embraces various fields of S&T. This is because advances in S&T in general are used to advantage for lowering the costs of sophisticated products, which will facilitate inclusive growth. In this regard, the AFIs listed in Table 1 attest to the use of progress in various areas of S&T for reducing costs. Therefore, research is needed in keeping abreast of the latest developments in science and engineering in general and later extracting ideas from this large pool that are relevant to the creation of a given AFI. The combing of such a wide swathe of scientific fields is essential to hone in on relevant portions of traditional sciences and/or emerging fields. An instance of an emerging field in engineering is

Biomimetics wherein nature-based frugal principles are being brought to bear on realizing products with judicious use of resources that perform effectively (Ball 2001).

Second, the leveraging of advances in S&T for creating AFIs would be incomplete without the rigor in subsequent design in view of the typical frugal amounts of resources involved. Therefore, a rigorous procedure should be established that minimizes the uncertainties involved in the design process. The minimization of uncertainties, related to aspects of design such as materials and manufacturing, would result in the best case scenario of lowering the *factor of safety* (*fos*) to ~ 1.5 . The *fos* of 1.5 is routinely employed in the aircraft industry (Shanley 1962, Norton 2006) along with rigorous design procedures to achieve robust aircrafts with lower weight for effective flight. Consequently, design of AFIs warrants research in systematizing a rigorous design methodology, accompanying a lower *fos*, that will lower various uncertainties thereby minimizing resource consumption. This entails R&D for identifying and later improving various aspects of design so as to minimize their uncertainties for achieving rigor with suitable trade-offs that do not hinder the functionality of the product (Otto and Antonsson 1991). Some of the aspects alluded to include the

following: accurate design and material models, near net-shape- and sustainable- manufacturing processes, various failure events including those with a lesser probability and extension of standardization of parts and other systems for inclusion of a wider range in shapes and sizes. Furthermore, such a rigorous design methodology should be applied iteratively to progressively iron out the various weak spots in a design that possibly escape notice in a single pass of the design process. Although this second element has been proposed from the perspective of engineering, the importance of rigor in scientific endeavors not based on the safety-factor approach is not lost out. Overall, rigor in all knowledge-based endeavors creating AFIs is imperative when avoiding excesses in the consumption of resources. Figure 1 shows a framework with flow between the first two elements for realizing an AFI.

And third, a strong nexus should be maintained with entities actively involved in developing *frugal innovations* for the lower income segment of society, to study their ingenious incorporation of thrift in grassroots products (Gupta 2012, 2013) and, where possible, facilitate its extension to AFIs. In fact such studies on the lower income segment should be undertaken in developed and developing countries alike to

avoid missing out on the next big idea. Although the content here can be subsumed under general research mentioned in the second element, its isolation as a third element is sought to highlight the significance of thrift in any *frugal innovation*, grassroots or the advanced type.

4.4. Relevance to Sustainable Development Although *frugal innovation* is characterized as a grassroots phenomenon that caters mainly to the bottom-of-the-pyramid, the all-inclusive applications of their sophisticated brethren are globally uplifting both the rich and poor. Some instances of this inclusivity include the ECG and ultrasound developed by GE, which have found wide spread applications in society at large (Immelt *et al.* 2009). The appearance of frugality in a range of advanced applications cannot be neglected and is a force for good as far as *sustainable development* is concerned. The conflicting goals of achieving better standards of living at both lower costs and minimal impact on the environment demand ever more rigorous involvement of scientific research on frugality, which is important for *sustainable development*.

4.5. Restraints on the Use of Frugality The “frugalization” of a product-design should be thoroughly vetted for its feasibility during conceptualization, especially in applications where the human life is at stake. Subsequently, the low-cost philosophy presented in this effort, with room for futuristic revisions, should be followed for the systematic realization of AFIs, whose performance is not sacrificed by their frugal nature.

5. CONCLUSIONS

This paper propounds a new class of *frugal innovation* called *Advanced Frugal Innovation* (AFI) that leverages developments in science

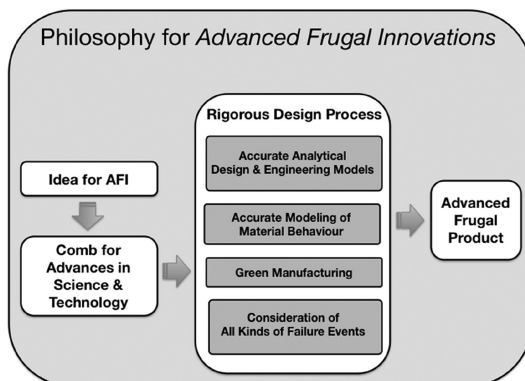


Figure 1 Low-Cost Philosophy for Realizing Advanced Frugal Innovations

and technology to achieve frugality in resource consumption. Accordingly, twenty-five *advanced frugal innovations* in sectors ranging from cars to healthcare, military-weaponry and particle physics have been presented. Most of these advanced innovations have been created by workers possessing advanced degrees beyond a bachelor's to

master the sophistication in technology required for their realization. A low cost philosophy using advances in science and technology has been shown to be necessary for the methodical design and creation of AFIs that though cheap are robust and effective in their performance. Such a philosophy would go a long way through

dedicated research and development activity for the creation of sophisticated products for the betterment of humans. Other than abetting the progress of *sustainable development*, AFIs have the potential to advance the boundaries of knowledge.

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